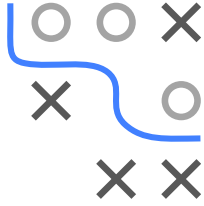
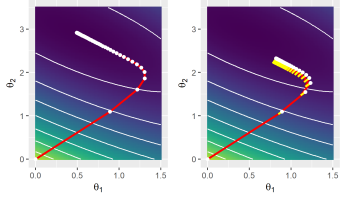
Supervised Learning



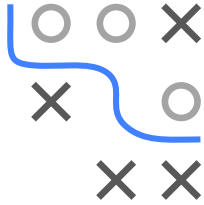
Regularization
Weight Decay and L2



Learning goals

- *L2* regularization with GD is equivalent to weight decay
- Understand how weight decay changes the optimization trajectory

WEIGHT DECAY VS. L2 REGULARIZATION I



Let's optimize *L2*-regularized risk of a model $f(\boldsymbol{x} \mid \boldsymbol{\theta})$

$$\min_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \min_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2$$

by GD. The gradient is

$$\nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \lambda \boldsymbol{\theta}$$

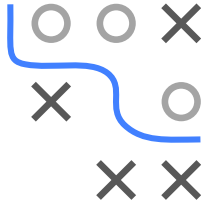
We iteratively update $\boldsymbol{\theta}$ by step size α times the negative gradient

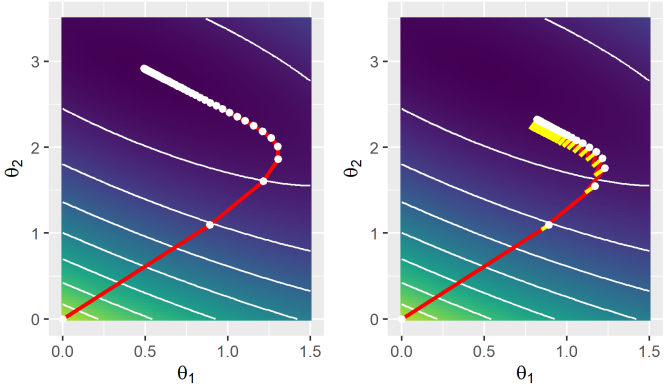
$$\begin{aligned} \boldsymbol{\theta}^{[\text{new}]} &= \boldsymbol{\theta}^{[\text{old}]} - \alpha \left(\nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]}) + \lambda \boldsymbol{\theta}^{[\text{old}]} \right) \\ &= \boldsymbol{\theta}^{[\text{old}]}(1 - \alpha\lambda) - \alpha \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]}) \end{aligned}$$

We see how $\boldsymbol{\theta}^{[old]}$ decays in magnitude – for small α and λ – before we do the gradient step. Performing the decay directly, under this name, is a very well-known technique in DL - and simply *L2* regularization in disguise (for GD).

WEIGHT DECAY VS. L2 REGULARIZATION II

In GD With WD, we slide down neg. gradients of $\mathcal{R}_{\text{emp}}$, but in every step, we are pulled back to origin.



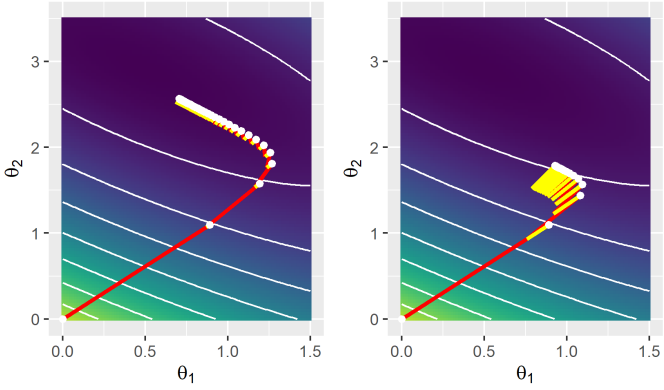


Without WD

With GD

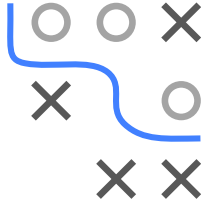
WEIGHT DECAY VS. L2 REGULARIZATION III

How strongly we are pulled back (for fixed α) depends on λ:



Small λ Large λ

CAVEAT AND OTHER OPTIMIZERS I



Caveat: Equivalence of weight decay and *L2* only holds for (S)GD!

- ▸ Hanson and Pratt 1988 originally define WD "decoupled" from gradient-updates $\alpha \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]})$ as $\boldsymbol{\theta}^{[\text{new}]} = \boldsymbol{\theta}^{[\text{old}]}(1 - \lambda') - \alpha \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}^{[\text{old}]})$
- This is equivalent to modern WD/*L2* (last slide) using reparameterization $\lambda' = \alpha\lambda$
- Consequence: if there is optimal λ', then optimal *L2* penalty is tightly coupled to α as $\lambda = \lambda'/\alpha$ (and vice versa)
- ▸ Loshchilov and Hutter 2019 show no equivalence of *L2* and WD possible for adaptive methods like Adam (Prop. 2)
- In many cases where SGD+*L2* works well, Adam+*L2* underperforms due to non-equivalence with WD
- They propose a variant of Adam decoupling WD from gradient updates (AdamW), increasing performance over Adam+*L2*